

Sumika Polymers NA

Date: 08/24/2022
For more information
or technical assistance
contact:

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Thermofil Circle[®] F615R99 Black

Easy Flow, Chemically Coupled 30% Glass Fiber-Reinforced Polypropylene Homopolymer containing 70% PCR Content in finished compound

Applications:

- Automotive and Industrial Parts Use providing superior impact, stiffness and dimensional stability
- Alternative to Long Glass-Fiber Filled PP, Glass-Filled Polyamide and Glass-Filled PBT

Physical Property	Test Value	ISO Test Method
Melt Flow	9 g/10 min	ISO 1133
Density	1.14 g/cc	ISO 1183
Flexural Modulus Flexural Strength	5800 MPa 102 MPa	ISO 178
Tensile Strength @ Yield Elongation @ Break	67 MPa 4.2 %	ISO 527
Notched Charpy Impact Strength @ 73°F (23°C) @ -22°F (-30°C)	10 kJ/m ² 6.5kJ/m ²	ISO 179/1eA
Rockwell Hardness (R Scale)	85	ISO 2039

“Preliminary Data Sheet”

Mechanical property testing performed on injection-molded specimens. v1

The nominal properties reported herein are typical of the product but do not reflect normal testing variance and therefore should not be used for specification purposes.

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Injection Molding Processing Recommendations

General Recommendations:

- Rear Temperature: 350 – 440°F (176 – 227°C)
- Middle, Front, and Nozzle Temps: 410 – 470°F (210 – 243°C)
- Stock Temperature 375 – 500°F (191 – 260°C)
- Mold Temperature: 100 – 160°F (38-71°C)
- Suggested shot size: 50 to 70%
- Maximize first stage injection pressure with fast fill rate preferred.
- Screw speed 30 to 60%
- Minimize the packing pressure.
- Back Pressure 50 to 100 psi.
- To achieve good dispersion of color concentrate it is recommended to use either a screw that has a mixing zone or a mixing nozzle.

Considerations:

The above conditions are general recommendations. Specific applications may require conditions outside of the given ranges. Melt temperatures above 500°F (260°C) may cause resin degradation and changes to resin properties. For additional assistance, contact your Sumika Polymers injection molding technical service representative.

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CRITICAL NOTICE:

Shrinkage Factors -

The shrinkage values provided in the table below were obtained in a laboratory setting with simple plaques and are reported as nominal values. They should be considered as estimated values only since resin shrinkage is influenced by several factors. The most common influencing factors include resin characteristics, part geometry, mold design, molding conditions, part temperature upon ejection from the mold and part cooling afterwards. Consequently, the values given below cannot be used as absolute.

Determining Shrinkage -

Since shrinkage is affected by so many factors, it is the user's responsibility to confirm the shrinkage for their own mold design and molding conditions before cutting tool steel. For the user to determine shrinkage, the suggested procedure is to finish only one core and cavity using these estimated values (staying steel safe). Based on the measurements of parts from this single core/cavity combination, the shrinkage factor can then be adjusted to fit the particular situation. A second iteration of mold changes (and costs) should be considered and applied after initial shrinkage factors are determined. If a similar tool is already present, run the material on this tool with normal process and handling conditions and determine the shrinkage. Remember to wait a full 72 hours for maximum shrinkage. Also, be aware the heat aging after molding can affect shrinkage.

For further assistance, please contact your account manager.

NOMINAL SHRINKAGE VALUES:

Sumika Polymers cannot warrant the shrinkage values reported below based on the notice given above.

The test results below are based on 150 x 90 x 3 mm plaques, edge gated, 1 mm thick gate.

Conditioning after molding:
48 hr at 23°C

Shrinkage:
0.2 – 1.0% (2 to 10 thousandths inch/inch)

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